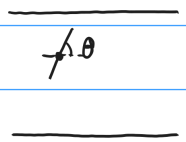
MATE 664 L21

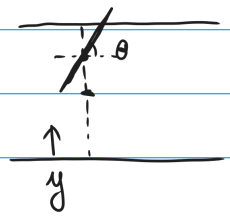
Monte Carlo Methods

Buffon's needle problem

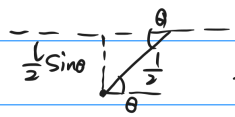


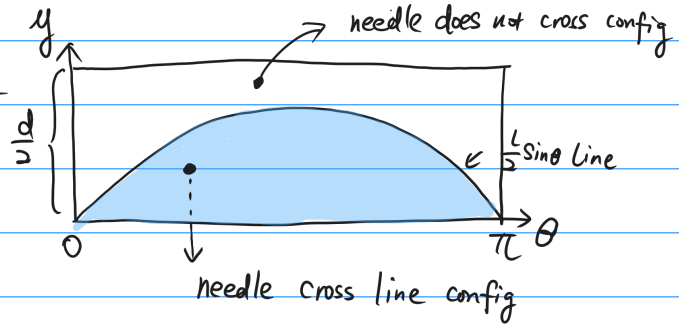
When does a needle length l cross parallel lines spacing d ?

Geometric intepretation



Each needle has 2 coordinate config
- center of needle y (from $\frac{d}{2} \to d$)
- rotation angle θ (from 0 to π)

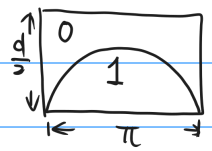




Formal request: find probability of $P(\text{cross})$ give θ, y

$$P = \frac{1}{\left(\frac{d}{2}\right)\cdot(\pi)} \cdot \int_0^{\pi} \frac{l}{2}\sin\theta \, d\theta = \frac{\frac{l}{2}\cdot(-\cos\theta)\Big|_0^{\pi}}{\frac{d}{2}\pi} = \frac{\frac{l}{2}\cdot 2}{\frac{d}{2}\pi} = \frac{2}{\pi}\frac{l}{d}$$

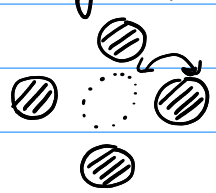
Or to think:



Find integral $P = \dfrac{\int_0^{\frac{d}{2}} \int_0^{\pi} f(y,\theta)\, dy\, d\theta}{L_y L_\theta}$

formally a Monte Carlo problem

Simulating diffusion by MC



swap probability $\Longleftrightarrow$ macroscopic flux

Problem to solve by Monte-Carlo method:

If measuring a property of A based on different configuration

$$\langle A \rangle_{ensemble} = \frac{\int_\Omega A(r) \exp(-\frac{E}{k_B T}) dr}{\int_\Omega \exp(-\frac{E}{k_B T}) dr}$$

It's the same as in the Buffon's problem!

General formula for MC-sampling

We want to estimate functional integral

$$F = \int_\Omega f(x)\, dx$$ with drawing from $\rho(x)$ distribution

$$F = \int_\Omega \boxed{\frac{f(x)}{\rho(x)}} \rho(x) dx \quad \& \quad \int_\Omega \rho(x) dx = 1$$

For Boltzmann Dynamics, the sampling is normalized in "partion function"

$$Z = \int_\Omega \exp(-\frac{E(r)}{k_B T}) dr$$ ($\vec{r}$ usually used for chemical coordinates)

$$\langle A \rangle = \frac{1}{Z} \int_\Omega A(r) \exp(-\frac{E(r)}{k_B T}) dr$$

$$= \int_\Omega \underbrace{A(r)}_{\text{Similar to } \frac{f}{\rho}} \underbrace{\frac{\exp(-\frac{E(r)}{k_B T})}{Z}}_{\text{normalized } \rho(r) \text{ distribution}} dr$$

In other words, Boltzmann Dynamics is trying to estimate a function $f(r) = A(r) \exp(-\frac{E}{k_B T})$

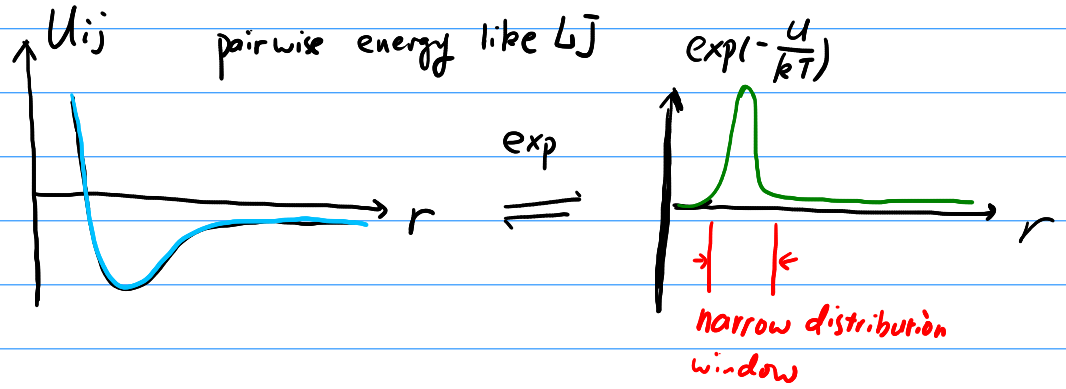
$$\rho(r) = \frac{\exp(-\frac{E(r)}{k_B T})}{Z}$$

Why uniform sampling does not work ?

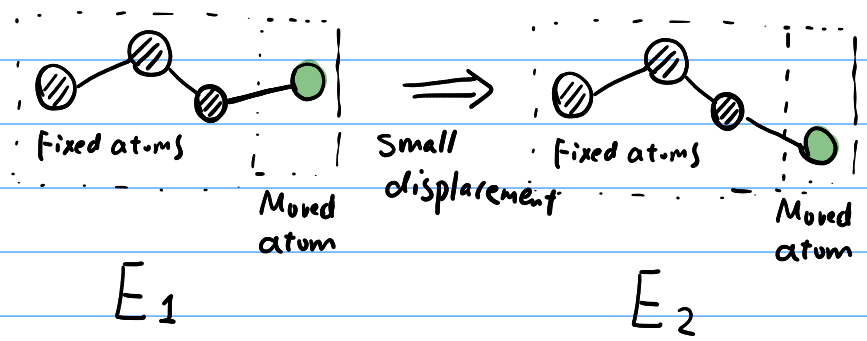
① From our previous (re)-formulation of Boltzmann Dynamics the distribution of property $A(r)$ is not uniform, but $A(r)\exp(-\frac{E}{k_BT})$

E: depends on the ensemble $\begin{pmatrix} NVT \rightarrow U \\ NPT \rightarrow G \end{pmatrix}$

② Uniform sampling mostly deals with systems with very low weights



Metropolis sampling (Metropolis MC)



If $\begin{cases} E_2 < E_1 \Rightarrow \text{accept configuration 2} \\ E_2 > E_1 \Rightarrow \text{accept with } p = \exp(-\frac{E_2 - E_1}{k_BT}) \end{cases}$

Metropolis MC is basically a thermodynamic sampling

$\Downarrow$

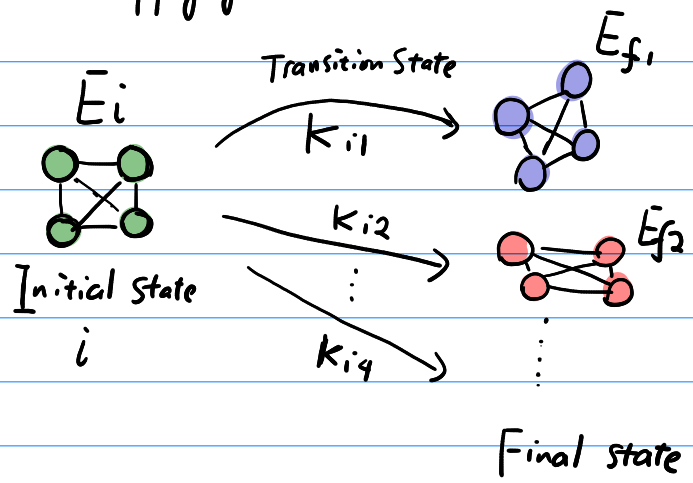
gives equilibrium configuration

$\Downarrow$

guaranteed to sample multiple energy wells

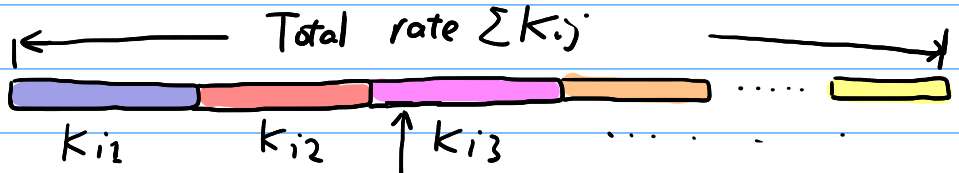
(But not essentially the "fastest" to visit)

KMC: applying Monte Carlo for kinetics



Instead of randomly switch states, KMC requires to get every transition state (TS) to estimate the total transition rates

rate constant k_{ij} is first order (unit s^{-1})



$k_{ij} = \exp\left(-\frac{\Delta E_{ij}}{k_B T}\right)$ Random number use R.N. to choose 1 process

Time scale? Random time increment of

$\Delta t = -\frac{1}{\sum k} \cdot \ln(RN_2)$ second R.N.